SIMATS ENGINEERING
Engineer to Excel

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO1 AT-3
TECHNICAL PRESENTATION

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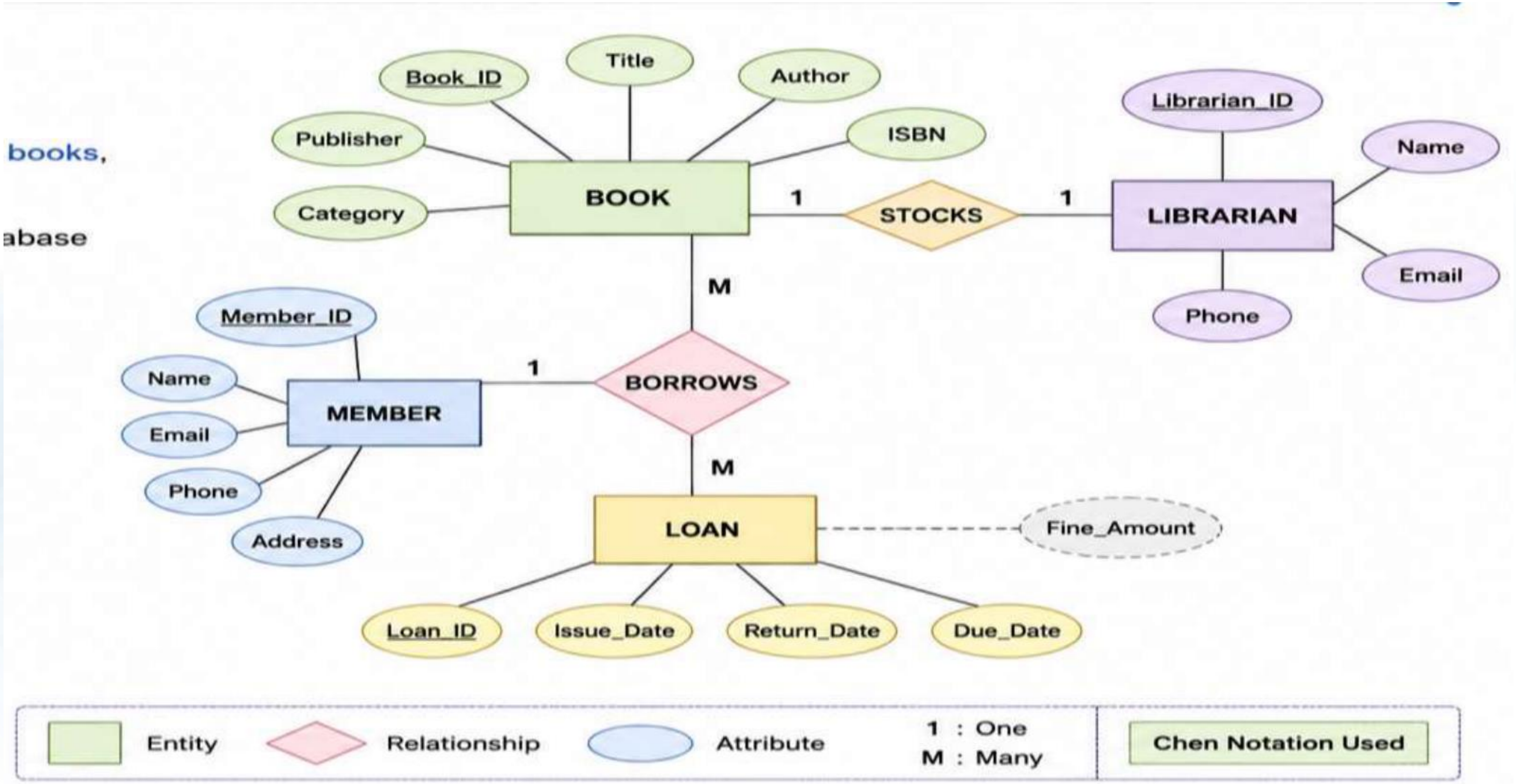
1) Create and present an ER diagram for an Online Library System, explaining each component clearly.

Introduction to Online Library System

- An Online Library System manages books, members, and borrowing activities.
- The ER diagram represents the database structure visually.
- It identifies entities, attributes, and relationships.
- It helps design an efficient and organized database.
- It minimizes data redundancy and improves data consistency.

ER Diagram for an Online Library System



Entities and Attributes

❖ **Main Entities**

- Book
- Member
- Librarian
- Loan

❖ **Key Attributes**

- Book: Book_ID, Title, Author, ISBN
- Member: Member_ID, Name, Email
- Librarian: Employee_ID, Name
- Loan: Loan_ID, Issue_Date, Return_Date

Relationships in ER Diagram

❖ **Relationships among Entities**

- A Member borrows many Books.
- A Book can be borrowed by different Members over time.
- A Librarian manages book issue and return.
- A Loan records borrowing transactions.
- Relationships define how entities interact.

MEMBER
|
Borrows
|
BOOK

BOOK
|
Recorded In
|
LOAN

Advantages and Complete ER Diagram

- **Benefits of ER Diagram**
- Simplifies database design.
- Reduces duplicate data.
- Improves data integrity.
- Makes future database modifications easier.
- Supports efficient query processing.

```
MEMBER -------- BORROWS --------
BOOK
   |                          |
   |                          |
   -------- LOAN -------- LIBRARIAN
```

Q2. Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.

Introduction to EER Model

❖ **Enhanced Entity Relationship (EER) Model**

- EER is an extension of the ER model.
- It supports more complex database designs.
- It introduces Generalization and Specialization.
- It models inheritance among entities.
- It represents real-world scenarios more accurately.

Generalization

- Bottom-up approach.
- Combines similar entities into one superclass.
- Common attributes are moved to the parent entity.
- Reduces redundancy.
- Improves database organization.

Professor
|
|
Administrator
|
Employee

Common Attributes:

- Employee_ID
- Name
- Salary

Specialization

- Top-down approach.
- Divides one entity into specialized subclasses.
- Each subclass has unique attributes.
- Provides better classification.

Supports inheritance.

```
            PERSON
               |
       ------------------
       |        |        |
Student Faculty Staff
```

Example Attributes:

Student → CGPA

Faculty → Subject

Staff → Department

Advantages and University Case Study

❖ **Advantages of EER Model**

- Represents complex relationships.
- Supports inheritance.
- Reduces data duplication.
- Improves flexibility.
- Suitable for large organizations like universities.

```
        PERSON
           |
 ------------------------
 |         |          |
Student  Faculty   Staff
 |         |          |
   Roll No  Subject
     Department
```

Conclusion

The EER model enhances the traditional ER model by introducing Generalization and Specialization, making university database design more flexible, organized, and efficient.

Q3. Present the concept of data independence (logical and physical) and its significance in DBMS architecture.

Introduction to Data Independence

❖ **Data Independence in DBMS**

- Data independence is the ability to modify one level of the database without affecting higher levels.
- It separates logical and physical data storage.
- It is achieved using the ANSI/SPARC three-level architecture.
- It improves flexibility and maintainability.
- It reduces the cost of database modifications.

External Level

|

Conceptual Level

|

Internal Level

Physical Data Independence

❖ **Physical Data Independence**

- Physical changes do not affect the conceptual schema.
- Storage devices or file organization can be changed.
- Improves storage efficiency.
- No changes are required in application programs.
- Supported by the Internal Level.

❖ **Examples**

- Changing HDD to SSD.
- Creating indexes.
- Compressing database files.

Applications

|

Conceptual Schema

|

Internal Schema

|

SSD / HDD

Logical Data Independence

- Logical changes do not affect user applications.
- New tables or attributes can be added.
- Existing applications continue working.
- Easier database expansion.
- Supported by the Conceptual Level.

❖ **Examples**

- Adding Email column.
- Creating a new Student table.
- Adding new relationships.

Student

ID
Name

↓

Student

ID
Name
Email

Significance of Data Independence

❖ **Importance of Data Independence**

- Reduces maintenance cost.
- Improves database flexibility.
- Supports future database growth.
- Protects application programs from changes.
- Makes database management more efficient.

Physical	Logical
Internal Level	Conceptual Level
Storage changes	Schema changes
Faster implementation	More difficult
No application changes	No user view changes

Q4. Deliver a technical presentation on the software components of a DBMS, explaining Query Processor, Storage Manager, and Transaction Manager.

Introduction to DBMS Components

❖ **Software Components of DBMS**

- DBMS consists of several software modules.
- Each module performs a specific function.
- Major components work together to process data.
- They ensure efficient data management.
- Three important components are Query Processor,
- Storage Manager, and Transaction Manager.

```
        User
          |
        DBMS
          |
----------------------
  |       |       |
  QP      SM      TM
```

(QP = Query Processor, SM = Storage Manager, TM = Transaction Manager)

Query Processor

- Accepts SQL queries from users.
- Checks syntax and semantics.
- Optimizes query execution.
- Generates an execution plan.
- Retrieves requested data efficiently.

❖ **Functions**

- SQL Parser
- Query Optimizer
- Execution Engine

SQL Query
|
Parser
|
Optimizer
|
Execution

Storage Manager

- Manages physical storage of data.
- Reads and writes data blocks.
- Controls indexes and files.
- Allocates disk space efficiently.
- Ensures secure data storage.

❖ **Components**

- File Manager
- Buffer Manager
- Authorization Manager

Mini Diagram

Database

Storage Manager

Files Buffers

Transaction Manager

- Controls database transactions.
- Ensures ACID properties.
- Handles commit and rollback operations.
- Maintains data consistency.
- Supports concurrent user access.

❖ **ACID Properties**

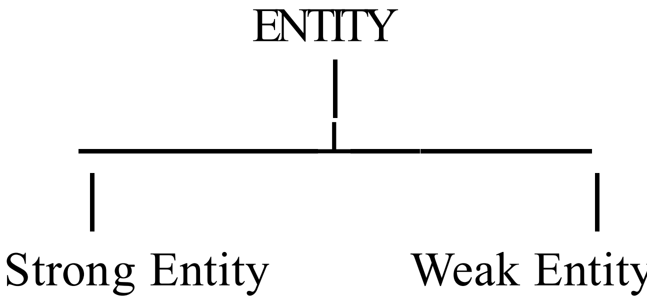
- Atomicity
- Consistency
- Isolation
- Durability

Start
|
Transaction
|
Success?
| |
Yes No
| |
Commit Rollback

Q5. Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.

Introduction to Strong and Weak Entities

- **Strong and Weak Entities in ER Modeling**
- An **Entity** represents a real-world object in a database.
- Entities are classified into **Strong** and **Weak** entities.
- Strong entities have their own primary key.
- Weak entities depend on strong entities for identification.
- ER diagrams use different symbols to represent them.



Strong Entity

- Has its own **Primary Key**.
- Exists independently in the database.
- Can be uniquely identified.
- Represented by a **single rectangle** in ER diagrams.
- Used as the parent entity in relationships.

Examples

- Student
- Employee
- Customer
- Product

STUDENT

| Student |

/ | \

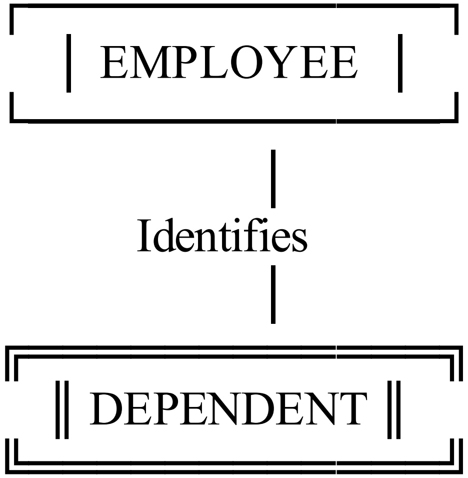
Student_ID Name
Email

Weak Entity

- Does **not** have a complete primary key.
- Depends on a strong entity for identification.
- Uses a **partial key (discriminator)**.
- Represented by a **double rectangle**.
- Connected through an **identifying relationship**.

Real-World Example

- Employee → Dependent



Dependent Attributes:

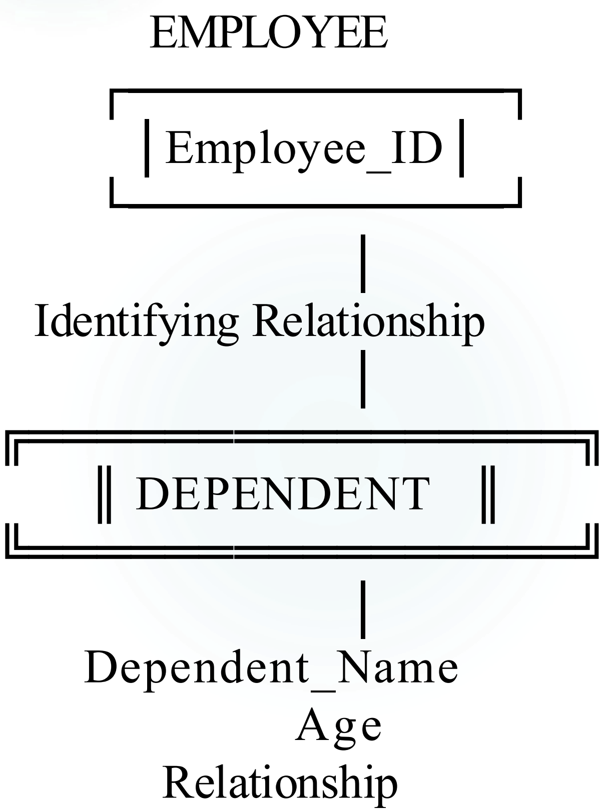
•Dependent_Name
•Age
•Relationship

Comparison of Strong and Weak Entities

Strong Entity	Weak Entity
Has Primary Key	No Primary Key
Independent	Depends on Strong Entity
Single Rectangle	Double Rectangle
Exists Alone	Cannot Exist Alone
Example: Student	Example: Dependent

Key Points:

- Strong entities uniquely identify records.
- Weak entities rely on owner entities.
- Both improve accurate database modeling.
- ER diagrams clearly distinguish both
- Entity types.



Conclusion

Strong entities can exist independently because they have a primary key, whereas weak entities depend on strong entities for identification. Together, they help create accurate and efficient ER models for real-world database systems.

THANK YOU